

Hydrodynamic-Aware Reinforcement Learning for Autonomous Marine Source Seeking

Nicola Forti*, Mattia Manneschi*, Irene Simonetti[†], and Giorgio Battistelli*

Abstract—Rapid localization of pollutant releases in coastal waters is complicated by time-varying currents, wind forcing, shoreline geometry, and mixing, which continuously deform the measured plume. We investigate a model-free, single-agent reinforcement-learning approach to marine source seeking in time-varying concentration fields generated with the MIKE 21 hydrodynamic model for the coastal area of Cecina, Italy. The task is formulated as a partially observable control problem and addressed by a centralized Maskable Proximal Policy Optimization (PPO) policy using concentration and displacement histories, wind and local-current information, and plume-memory features. Spatial sensing is varied among central-agent-only measurements and ideal co-moving formations with one or two sensing rings. The three policies are compared with a field-climbing baseline across vehicle speeds, wind-forcing conditions, and plume ages.

Index Terms—marine robotics, source seeking, pollutant plume tracing, reinforcement learning, formation sensing, hydrodynamic modeling

I. INTRODUCTION

Locating an unknown pollutant source in coastal or port waters is a time-critical task in environmental monitoring, emergency response, and maritime safety. Autonomous underwater and surface vehicles can acquire in-situ measurements without exposing operators to hazardous substances. Marine source seeking nevertheless remains challenging because currents, tides, wind-driven circulation, turbulent mixing, and interactions with coastal boundaries continuously transport and deform the plume, making the instantaneous concentration gradient not necessarily aligned with the direction of the source. Chemical plume tracing has been demonstrated using reactive, bio-inspired, and model-based guidance [1]–[3]. Spatially separated measurements can improve local field observability [4]–[6]. Source estimation, information-driven motion planning, and cooperative search have also been investigated in spatial and spatio-temporal fields [7]–[10]. These approaches provide valuable analytical or training-free alternatives. Gradient-based controllers can nevertheless be unreliable in advected plumes, whereas model-based active methods require an explicit online representation of the source and field evolution.

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Reinforcement learning (RL) instead learns a closed-loop strategy through repeated interaction with simulated fields. Proximal Policy Optimization (PPO) provides stable clipped policy updates [11] and reliable implementations [12]. Because source localization is partially observable, existing RL methods use particle-filter features [13], recurrent policies [14], or source-belief convergence for autonomous stopping [15]. Recent physics-informed multi-agent RL approaches further address scalable distributed control [16]. However, the effects of temporal memory, spatial sensing geometry, and vehicle speed remain insufficiently characterized using high-resolution, time-dependent coastal plume simulations.

This paper studies these effects using pollutant-transport simulations of the coastal area of Cecina, Italy. The hydrodynamic model generates offline training and evaluation fields and is not solved onboard. At execution time, a centralized policy controls a ground-referenced central agent using local concentration measurements, finite histories of concentration and displacement, time-aligned wind and current vectors, and plume-memory features. We compare three spatial sensing configurations: concentration measurements acquired only by the central agent, a fixed formation comprising the central agent and one ring of eight sensing agents at a radius of 20 m, and a formation with two sensing rings at radii of 20 and 50 m. The agents on the rings are modeled as ideal co-moving sensing platforms whose concentration measurements are made available to the central policy. The main contributions are:

- a POMDP formulation defining latent hydrodynamic states, locally available observations, finite histories, and privileged source information;
- a systematic comparison of three MaskablePPO sensing configurations with a Field Climbing Method (FCM) baseline across multiple vehicle-speed limits;
- an evaluation across forcing conditions and plume ages using success rate and mean steps to the source neighborhood conditional on success.

The remainder of the paper is organized as follows. Section II formulates the source-seeking problem, Section III presents the hydrodynamic dataset and PPO controller, Section IV reports the experimental results, and Section V concludes the paper.

II. PROBLEM FORMULATION

A. POMDP formulation of source seeking

Consider a central agent tasked with locating a stationary but unknown source at p_s within a navigable domain $\mathcal{D} \subset$

$\mathbb{R}^2$. Let Ξ denote the set of hydrodynamic realizations. For each source position $\mathbf{p}_s \in \mathcal{D}$ and hydrodynamic realization $\xi \in \Xi$, let $c(\cdot, \cdot; \mathbf{p}_s, \xi) : \mathcal{D} \times [0, T_{\text{sim}}] \rightarrow \mathbb{R}_{\geq 0}$ denote the resulting depth-averaged pollutant concentration field. Thus, $c(\mathbf{p}, t; \mathbf{p}_s, \xi)$ is the concentration at position $\mathbf{p}$ and time t , expressed in g/m^3 . Each hydrodynamic realization $\xi \in \Xi$ indexes a possible evolution of the marine environment and is associated with the corresponding depth-averaged concentration and current fields $c(\mathbf{p}, t; \mathbf{p}_s, \xi)$ and $\mathbf{u}_\xi(\mathbf{p}, t)$, together with the wind forcing $\mathbf{w}_\xi(t)$.

A pointwise concentration sensor located at $\mathbf{p}$ is described by the measurement model

$$y(\mathbf{p}, t) = c(\mathbf{p}, t; \mathbf{p}_s, \xi) + \varepsilon(\mathbf{p}, t), \quad (1)$$

where ε represents measurement uncertainty.

Let $\mathcal{Q}$ denote the set of admissible sensing configurations, and let $q \in \mathcal{Q}$ index a generic configuration. Conditional on q , the source-seeking problem is modeled as the finite-horizon partially observable Markov decision process (POMDP) [17]

$$\mathcal{P}_q = \langle \mathcal{X}, \mathcal{A}, \kappa, \mathcal{Y}_q, o_q, r, \mu_0, \gamma, K_{\text{max}} \rangle, \quad (2)$$

where $\mathcal{X}$ and $\mathcal{A}$ are the state and nominal action spaces, κ is the state-transition kernel, $\mathcal{Y}_q$ and o_q are the configuration-dependent observation space and observation kernel, r is the transition-reward function, μ_0 is the initial-state distribution, $\gamma \in (0, 1]$ is the discount factor, and K_{max} is the episode horizon.

Let $\mathcal{E}$ denote the latent environmental-state space. At decision time k , $\eta_k \in \mathcal{E}$ denotes the latent environmental state. This state comprises the concentration and hydrodynamic fields, the wind forcing, the physical simulation time, and any additional variables required to describe their evolution. Each hydrodynamic realization $\xi \in \Xi$ is associated with a possible realization of the evolution of η_k and of the corresponding concentration, current, and wind fields.

The complete latent state of the agent-environment system is $\mathbf{x}_k = (\mathbf{p}_k, \mathbf{p}_s, \eta_k) \in \mathcal{X} \subseteq \mathcal{D} \times \mathcal{D} \times \mathcal{E}$, where $\mathbf{p}_k$ is the central-agent position at decision time k . The source position remains fixed within an episode, while the agent position and the environmental state evolve. State evolution satisfies $\mathbf{x}_{k+1} \sim \kappa(\cdot | \mathbf{x}_k, a_k)$, where κ combines the controlled agent motion with the evolution of the environmental state.

The central agent is modeled as a planar kinematic point controlled at intervals of $\Delta t = 10$ s. At decision time $t_k = t_0 + k\Delta t$, action $a_k = (j_k, v_k) \in \mathcal{A}$ selects a heading index $j_k \in \{1, \dots, 8\}$ and a positive speed v_k . Its ground-referenced position evolves according to

$$\mathbf{p}_{k+1} = f(\mathbf{p}_k, a_k) \triangleq \mathbf{p}_k + \Delta t v_k \mathbf{e}_{j_k}, \quad (3)$$

where $\mathbf{e}_j = [\cos((j-1)\pi/4) \ \sin((j-1)\pi/4)]^\top$. For a specified maximum speed v_{max} , the discrete action space is $\mathcal{A} = \{1, \dots, 8\} \times \mathcal{V}(v_{\text{max}})$, where $\mathcal{V}(v_{\text{max}}) = \left\{ \frac{l}{K_v} v_{\text{max}} : l = 1, \dots, K_v \right\}$ and $K_v = 5$. Thus, $\mathcal{A}$ contains $8K_v = 40$ heading-speed combinations. Since $\|\mathbf{e}_j\| = 1$, axial and diagonal actions produce the same displacement

magnitude at a given speed. To enforce coastline and domain constraints, the environment constructs the state-dependent admissible-action set $\mathcal{A}_k = \{a \in \mathcal{A} : f(\mathbf{p}_k, a) \in \mathcal{D}\}$. Its binary representation is the action-validity mask $m_k : \mathcal{A} \rightarrow \{0, 1\}$ defined by $m_k(a) = \mathbf{1}_{\{a \in \mathcal{A}_k\}}$. The mask is a deterministic feasibility signal computed from the current position and navigation map and supplied to the controller before action selection.

At time k , the agent receives the configuration-dependent observation $\mathbf{y}_k^q \sim o_q(\cdot | \mathbf{x}_k)$. The vector $\mathbf{y}_k^q$ contains the pointwise concentration measurements available under configuration q together with the local wind and current vectors. From the current observation, selected components of its finite history, the realized displacement history, and the recursively updated plume-memory vector $\mathbf{h}_k$, the deterministic feature map ϕ_q constructs the policy input

$$\mathbf{z}_k^q = \phi_q(\mathbf{y}_{k-H:k}^q, \Delta \mathbf{p}_{k-H:k-1}, \mathbf{h}_k) \in \mathcal{Z}_q, \quad (4)$$

where $\Delta \mathbf{p}_n = \mathbf{p}_{n+1} - \mathbf{p}_n$ for $n = k-H, \dots, k-1$. The source position and the complete environmental fields are not included in $\mathbf{z}_k^q$.

To solve each POMDP, we use an action-masked stochastic actor trained with Proximal Policy Optimization (PPO) [11]. Let $\Delta(\mathcal{A})$ denote the probability simplex over $\mathcal{A}$. For sensing configuration q , the unmasked actor parameterizes the stochastic policy $\pi_{\theta_q} : \mathcal{Z}_q \rightarrow \Delta(\mathcal{A})$. For any policy input $\mathbf{z} \in \mathcal{Z}_q$, $\pi_{\theta_q}(a | \mathbf{z})$ denotes the probability of selecting action $a \in \mathcal{A}$ and satisfies $\pi_{\theta_q}(a | \mathbf{z}) \geq 0$, $\sum_{a \in \mathcal{A}} \pi_{\theta_q}(a | \mathbf{z}) = 1$. Renormalizing this distribution over $\mathcal{A}_k$ gives the masked policy

$$\pi_{\theta_q}^m(a | \mathbf{z}_k^q, m_k) = \frac{m_k(a) \pi_{\theta_q}(a | \mathbf{z}_k^q)}{\sum_{a' \in \mathcal{A}} m_k(a') \pi_{\theta_q}(a' | \mathbf{z}_k^q)}. \quad (5)$$

The target set is the source-centered disk $\mathcal{G}_s = \{\mathbf{p} \in \mathcal{D} : \|\mathbf{p} - \mathbf{p}_s\| \leq r_s\}$, with $r_s = 50$ m. The episode horizon is $K_{\text{max}} = 1080$ decision steps, corresponding to $K_{\text{max}} \Delta t = 3$ h of simulated time. The first hitting time of $\mathcal{G}_s$ is $\tau_s = \inf \{k \geq 1 : [\mathbf{p}_{k-1}, \mathbf{p}_k] \cap \mathcal{G}_s \neq \emptyset\}$, where $[\mathbf{p}_{k-1}, \mathbf{p}_k]$ is the closed motion segment between consecutive positions.

Let $\mathbb{P}_q^{\theta_q}$ and $\mathbb{E}_q^{\theta_q}$ denote the trajectory probability measure and expectation induced by μ_0 , κ , o_q , and $\pi_{\theta_q}^m$. Then the RL objective can be written as

$$J_q(\theta_q) = \mathbb{E}_q^{\theta_q} \left[\sum_{k=0}^{K-1} \gamma^k r(\mathbf{x}_k, a_k, \mathbf{x}_{k+1}) \right], \quad (6)$$

where $K = \min\{\tau_s, K_{\text{max}}\}$. The masked PPO update used to approximately maximize (6) is described in Sec. III. Although the true source position enters the training reward and the simulated success condition, it is excluded from the policy input $\mathbf{z}_k^q$.

B. Sensing configurations and policy input

The sensing configurations are indexed by $\mathcal{Q} = \{0, 1, 2\}$ and specified by the active-ring sets $\mathcal{S}_0 = \emptyset$, $\mathcal{S}_1 = \{1\}$,

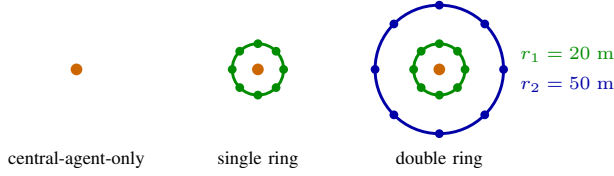


Fig. 1. Centralized concentration-sensing geometries. Orange markers denote the central agent; ring markers denote ideal co-moving sensing agents.

and $\mathcal{S}_2 = \{1, 2\}$, corresponding to central-agent-only, single-ring, and double-ring sensing, respectively. Their geometries are illustrated in Fig. 1.

Using (1) and suppressing the dependence of the concentration field on $\mathbf{p}_s$ and ξ , the central concentration measurement is $y_k = c(\mathbf{p}_k, t_k) + \varepsilon_k$, where $\varepsilon_k = \varepsilon(\mathbf{p}_k, t_k)$. For each active ring $\ell \in \mathcal{S}_q$ and direction $j \in \{1, \dots, 8\}$, the sensing position is $\mathbf{p}_k^{\ell,j} = \mathbf{p}_k + r_\ell \mathbf{e}_j$, with $r_1 = 20$ m and $r_2 = 50$ m. The offsets are aligned with the fixed global directions in (3). The ring-measurement vector $\mathbf{y}_k^\ell \in \mathbb{R}^8$ has components $[\mathbf{y}_k^\ell]_j = c(\mathbf{p}_k^{\ell,j}, t_k) + \varepsilon_k^{\ell,j}$, where $\varepsilon_k^{\ell,j} = \varepsilon(\mathbf{p}_k^{\ell,j}, t_k)$. The reported experiments are noise-free, i.e., $\varepsilon(\mathbf{p}, t) \equiv 0$.

Ring locations are modeled as ideal co-moving sensing agents whose measurements are instantaneously available to the central policy. Measurements at locations lying on land or outside the simulated domain are treated as unavailable and zero-filled in the policy input. These virtual agents have no independent dynamics, policies, or control actions. Therefore, the sensing configuration modifies only the observation model and does not change the single-agent nature of the decision process.

Let $H = 9$ denote the number of previous samples retained, and let $\text{col}(\cdot)$ denote vector concatenation. For configuration q , the ring-measurement block is $\tilde{\mathbf{y}}_k^q \triangleq \text{col}_{\ell \in \mathcal{S}_q}(\mathbf{y}_k^\ell, \mathbf{y}_{k-H:k-1}^\ell)$. Each active ring contributes $8(H+1)$ components, and therefore $\tilde{\mathbf{y}}_k^q \in \mathbb{R}^{8(H+1)|\mathcal{S}_q|}$, with $\tilde{\mathbf{y}}_k^0$ defined as the empty vector.

Consistently with (4), ϕ_q constructs $\mathbf{z}_k^q$ by concatenating the current central measurement y_k , its history $y_{k-H:k-1}$, the realized displacement history $\Delta \mathbf{p}_{k-H:k-1}$, the wind and current vectors $\mathbf{w}_k, \mathbf{u}_k \in \mathbb{R}^2$, the plume-memory vector $\mathbf{h}_k$, and $\tilde{\mathbf{y}}_k^q$. The current $\mathbf{u}_k$ is evaluated at the central-agent position and time t_k , whereas the spatially uniform wind $\mathbf{w}_k$ is synchronized at t_k .

The plume-memory vector is $\mathbf{h}_k = [\hat{\mathbf{p}}_k^{\max} - \mathbf{p}_k, y_k^{\max}, \zeta_k/K_{\max}] \in \mathbb{R}^4$, where $y_k^{\max} = \max\{y_{k-1}^{\max}, y_k\}$, with $y_0^{\max} = y_0$, and $\hat{\mathbf{p}}_k^{\max}$ is a central-agent position at which this maximum was observed. The counter ζ_k records the number of steps since y_k last exceeded the plume-detection threshold $y_{\text{th}} = 0.5$ g/m³; it is initialized at zero, reset when $y_k > y_{\text{th}}$, and incremented otherwise. The same numerical threshold is applied to the simulated concentration field when selecting plume-positive initial cells. The central-agent-only, single-ring, and double-ring policy inputs contain 36, 116, and 196 components, respectively.

C. Reward function

For the transition $(\mathbf{x}_k, a_k, \mathbf{x}_{k+1})$, we define the source distance, distance progress, and variation of the central concentration measurement as $d_k = \|\mathbf{p}_k - \mathbf{p}_s\|_2$, $\Delta d_{k+1} = d_k - d_{k+1}$, $\Delta y_{k+1} = y_{k+1} - y_k$. Let d_{k+1} denote the distance from $\mathbf{p}_{k+1}$ to land, let $I_{k+1} = \mathbf{1}_{\{\mathbf{p}_k, \mathbf{p}_{k+1}\} \cap \mathcal{G}_s \neq \emptyset\}}$ denote source capture, and define the normalized remaining horizon at decision step k as $\chi_k \triangleq 1 - k/K_{\max}$, $k \in \{0, \dots, K_{\max}\}$. The concentration-shaping function is defined as $\varphi_y(\Delta y) = \begin{cases} +\alpha_y, & \Delta y > 0, \\ -\alpha_y, & \Delta y \leq 0. \end{cases}$. The transition reward $r_{k+1} \triangleq r(\mathbf{x}_k, a_k, \mathbf{x}_{k+1})$ is computed as

$$r_{k+1} = I_{k+1}(\alpha_g + \alpha_e \chi_{k+1}) + \alpha_d \Delta d_{k+1} + \varphi_y(\Delta y_{k+1}) - \alpha_t - \alpha_L \left(1 - \frac{\tilde{d}_{k+1}}{\tilde{d}_0}\right) \mathbf{1}_{\{\tilde{d}_{k+1} \in (0, \tilde{d}_0)\}}. \quad (7)$$

where: $\alpha_g = 100$, $\alpha_e = 50$, $\alpha_d = 100/(3000 \text{ m})$, $\alpha_y = 0.05$, $\alpha_t = 0.1$, $\alpha_L = 15$, and $\tilde{d}_0 = 30$ m. The first term in (7) rewards source capture and favors earlier successful termination, the second rewards progress toward the source, and φ_y rewards a strict increase in the central concentration measurement between consecutive time steps. This term depends on the temporal concentration variation measured along the central-agent trajectory and does not represent a spatial concentration gradient. The remaining terms penalize elapsed time and positions close to land.

III. HYDRODYNAMIC-AWARE PPO

A. Hydrodynamic data

The dataset was generated using the MIKE 21 Flow Model FM [18], which solves the depth-averaged shallow-water equations under hydrostatic and Boussinesq assumptions by finite-volume discretization on an unstructured triangular mesh. Pollutant transport is described by an Eulerian advection-dispersion equation for a passive, nonreactive tracer.

The computational domain covers approximately 16×14 km around the port of Cecina, Italy, with mesh resolution varying from about 10 m in the area of interest to 150 m near the outer boundaries. For RL training and evaluation, concentration and current fields are extracted over the UTM Zone 32N box $[619000, 622000] \times [4794500, 4797000]$ m, covering a 3×2.5 km coastal area, and represented on a 300×250 grid with nominal 10 m spacing.

Hourly currents and free-surface elevations from the Copernicus Marine Service Mediterranean product are imposed along the three open boundaries using Flather-type conditions [19], [20], providing dynamical downscaling from approximately 4.5 km to the local coastal resolution. Spatially uniform, time-varying wind forcing is obtained from the ERA5 10-m wind components [21]. Wave-induced processes and freshwater inflows are not modeled.

To select the open-boundary forcing, the 2024-2025 CMEMS series was partitioned into 48-h windows and grouped by k -means using mean current intensity, directional

TABLE I
WIND-FORCING CLASSES.

Class	Direction sector	Speed (m/s)	Regime
V0	no dominant sector	1.5–2	calm/weak
V1	NW, 310°–330°	8–10	persistent
V2	SW, 230°–250°	8–10	persistent
V3	E, 110°–130°	6–8	persistent

persistence, and free-surface variability. All simulations use the representative 19-20 May 2024 window, characterized by moderate, directionally persistent currents and low free-surface variability. This regime is combined with four ERA5-derived wind classes $\mathcal{F} = \{V0, V1, V2, V3\}$, summarized in Table I.

Each 48-h wind profile comprises 8 h below 2 m/s, a 6-h ramp to its class-dependent maximum, 12 h at maximum intensity, a 6-h decay, and weak winds for the remaining time. Wind direction varies within the corresponding 20° sector, whereas V0 has no dominant direction.

Let $\mathcal{I}_{\text{src}} = \{1, \dots, N_{\text{src}}\}$, with $N_{\text{src}} = 132$, index source locations spaced approximately 200 m apart. Each source releases the passive tracer with discharge $Q_{\text{src}} = 50$ L/s, concentration $c_{\text{src}} = 1000$ g/m³, and initial discharge velocity $\mathbf{v}_{\text{src}} = (-0.5, -0.5)$ m/s. For $(s, \nu) \in \mathcal{I}_{\text{src}} \times \mathcal{F}$, the corresponding concentration field is $c^{s, \nu}(\mathbf{p}, t) \triangleq c(\mathbf{p}, t; \mathbf{p}_s, \boldsymbol{\xi}_\nu)$, where $\boldsymbol{\xi}_\nu$ denotes the forcing conditions of class ν . The dataset therefore contains $N_{\text{src}}|\mathcal{F}| = 528$ source-forcing combinations. For each $\nu \in \mathcal{F}$, the current field $\mathbf{u}^\nu(\mathbf{p}, t)$ is obtained from a source-free reference simulation, assuming that source discharge negligibly affects the hydrodynamics beyond the immediate release region. Concentration and current outputs are stored in netCDF format on the 10-m grid at 2 min intervals.

For normalized plume age β , define $\tau(\beta) = \beta T_{\text{sim}}$, where $T_{\text{sim}} = 48$ h. The experiments use $\beta \in \mathcal{B} = \{1/4, 1/2, 3/4\}$, corresponding to $\tau(\beta) \in \{12, 24, 36\}$ h, and initialize the simulation at $t_0(\beta) = t_{\text{start}} + \tau(\beta)$. An episode scenario is indexed by $(s, \nu, \beta) \in \mathcal{I}_{\text{src}} \times \mathcal{F} \times \mathcal{B}$. At control step k , $t_k = t_0(\beta) + k\Delta t$, and the environment uses the synchronized fields $c^{s, \nu}(\mathbf{p}, t_k)$, $\mathbf{u}^\nu(\mathbf{p}, t_k)$, and $\mathbf{w}^\nu(t_k)$. Stored outputs are linearly interpolated at the $\Delta t = 10$ s control interval. The policy receives $\mathbf{w}_k = \mathbf{w}^\nu(t_k)$ and $\mathbf{u}_k = \mathbf{u}^\nu(\mathbf{p}_k, t_k)$, where the current is evaluated at the central-agent position and the wind is spatially uniform. Episodes sharing (s, ν, β) may use different initial positions $\mathbf{p}_0$.

B. Policy architecture and PPO updates

For each sensing configuration $q \in \mathcal{Q}$, we train an actor-critic pair. The actor network, parameterized by θ_q , represents the unmasked policy π_{θ_q} and, after applying the action-validity mask in (5), defines the distribution used to select the action. The critic $V_{\psi_q} : \mathcal{Z}_q \rightarrow \mathbb{R}$ approximates the conditional expected discounted return from the current policy input and is used only during training. The actor and critic are represented by separate multilayer perceptrons with two 256-unit hidden layers and hyperbolic-tangent activations. The actor produces

TABLE II
PPO HYPERPARAMETERS.

Parameter	Value	Parameter	Value
Hidden layers	2×256 , tanh	n_{env}	6
$n_{\text{steps}}/\text{env.}$	4096	Rollout size	24576
Minibatch size	128	Epochs/update	10
Discount γ	0.99	GAE λ	0.95
Clip ϵ	0.15	c_V / c_H	0.3 / 0.01
Max. grad. norm	0.5	Target KL	0.02
Obs./reward norm.	yes/no	Action outputs	40

logits for the $|\mathcal{A}| = 40$ actions, whereas the critic produces a scalar value estimate. Policy inputs are standardized using means and variances estimated during training and held fixed during evaluation. Rewards retain the scale defined in (7).

During each rollout, transitions are generated using fixed actor and critic parameters, denoted by θ_q^{old} and ψ_q^{old} . Let b_{k+1} be the continuation indicator, equal to one for a nonterminal transition and zero otherwise. The advantage of action a_k measures its expected return relative to the critic estimate at $\mathbf{z}_k^q$. Starting from the final transition of each rollout, generalized advantage estimation (GAE) [22] computes its estimate $\hat{A}_k$ recursively as

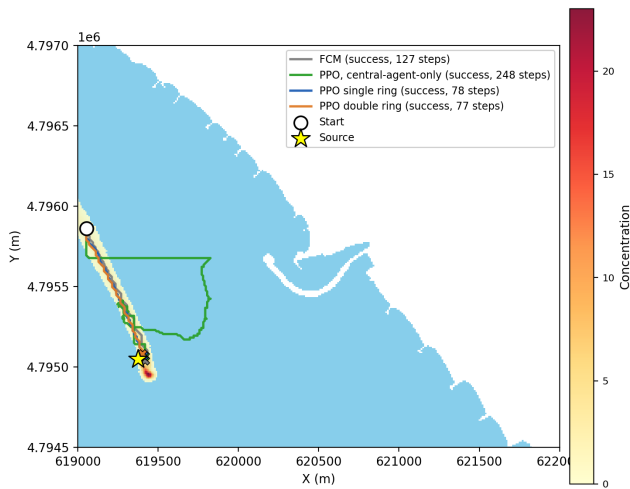
$$\begin{aligned} \delta_k &= r_{k+1} + \gamma b_{k+1} V_{\psi_q^{\text{old}}}(\mathbf{z}_{k+1}^q) - V_{\psi_q^{\text{old}}}(\mathbf{z}_k^q), \\ \hat{A}_k &= \delta_k + \gamma \lambda b_{k+1} \hat{A}_{k+1}, \end{aligned} \quad (8)$$

where δ_k is the temporal-difference residual and $\lambda = 0.95$. The corresponding target for critic regression is $\hat{G}_k = \hat{A}_k + V_{\psi_q^{\text{old}}}(\mathbf{z}_k^q)$. During a PPO update, θ_q^{old} remains fixed. The probability ratio $\rho_k(\theta_q) = \frac{\pi_{\theta_q}^m(a_k | \mathbf{z}_k^q, m_k)}{\pi_{\theta_q^{\text{old}}}^m(a_k | \mathbf{z}_k^q, m_k)}$ measures the change in the probability assigned to the sampled action by the updated actor relative to the actor that generated the rollout. PPO maximizes the clipped policy objective [11]

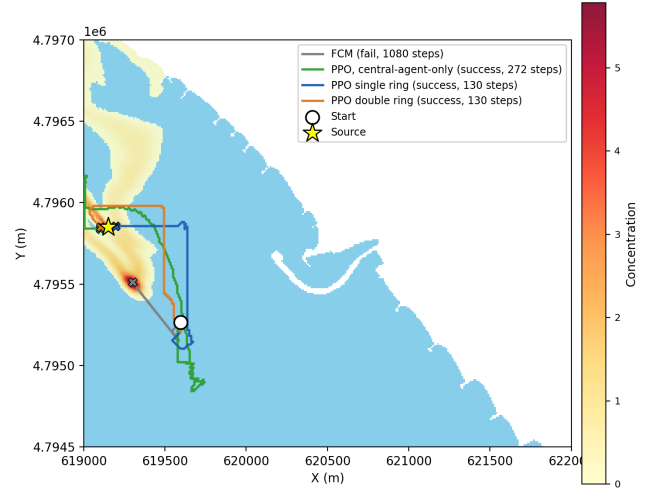
$$\bar{\mathcal{L}}_q(\theta_q) = \mathbb{E}_k \left[\min \left\{ \rho_k(\theta_q) \hat{A}_k, \tilde{\rho}_k(\theta_q) \hat{A}_k \right\} \right], \quad (9)$$

where $\tilde{\rho}_k(\theta_q) \triangleq \text{clip}(\rho_k(\theta_q), 1 - \epsilon, 1 + \epsilon)$ is the clipped probability ratio, $\mathbb{E}_k$ denotes the empirical average over sampled rollout transitions, and $\epsilon = 0.15$. The joint actor-critic objective is $\mathcal{J}_q^{\text{PPO}}(\theta_q, \psi_q) = \bar{\mathcal{L}}_q(\theta_q) - c_V \mathcal{L}_q^V(\psi_q) + c_H \mathcal{H}_q(\theta_q)$, where $\mathcal{L}_q^V(\psi_q) = \mathbb{E}_k[(V_{\psi_q}(\mathbf{z}_k^q) - \hat{G}_k)^2]$ is the critic regression loss and $\mathcal{H}_q(\theta_q)$ is the mean entropy of the masked policy. The implementation uses MaskablePPO and Stable-Baselines3 [12]. The PPO hyperparameters are listed in Table II. Training uses six parallel environments, two for each plume age $\beta \in \mathcal{B}$. Each rollout therefore contains $6 \times 4096 = 24576$ transitions, which are shuffled into minibatches of 128 and processed for up to ten epochs. Policy updates terminate early when the approximate Kullback-Leibler divergence exceeds the prescribed target.

The single-ring PPO policy is trained for 6×10^6 environment transitions and used as the parent checkpoint. The central-agent-only and double-ring PPO policies are initialized from this checkpoint by resizing the actor and critic input



(a) Favorable field-climbing case: $V0$, $\beta = 3/4$.



(b) Challenging field-climbing case: $V2$, $\beta = 1/2$.

Fig. 2. Representative held-out trajectories at $v_{\max} = 1.2$ m/s. Panel (a) illustrates a favorable field-climbing scenario: FCM follows the plume efficiently, while central-agent-only PPO takes a substantial detour, consistent with the limited instantaneous directional information provided by a single pointwise concentration measurement. In the more challenging scenario of panel (b), FCM becomes trapped at a local maximum of the displayed final concentration field and times out, whereas all PPO variants reach the source.

layers, and each is fine-tuned for an additional 2×10^6 transitions.

IV. EXPERIMENTAL RESULTS

The 132 source locations are partitioned into disjoint sets of 106 training locations and 26 held-out evaluation locations. For each value of $v_{\max}$, every method is evaluated on the 26 held-out source locations under all four forcing classes $\nu \in \mathcal{F}$ and all three initial plume ages $\beta \in \mathcal{B}$, with five episodes for each source-forcing-age combination. This yields $26 \times 4 \times 3 \times 5 = 1560$ evaluation episodes per method and speed limit. Initial positions are restricted to detected-plume cells satisfying $c(\mathbf{p}_0, t_0) > c_{\text{sp}}$, with $c_{\text{sp}} = 0.5$ g/m³. The initial source distance is sampled uniformly from an interval contained in $[300, 3000]$ m, and the nearest valid plume cell is selected.

At each step, the training-free FCM baseline estimates the local concentration gradient by solving a least-squares problem using the central measurement and an eight-point sensing ring, consistently with formation-based gradient estimation [6]. The gradient estimate is processed using the first- and second-moment updates of Adam, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon_A = 10^{-8}$ [23]. For each tested value of $v_{\max}$, the nominal Adam step parameter is set to $\alpha_{\text{FCM}} = v_{\max} \Delta t$.

Figure 2 provides qualitative examples of the controller behavior in two held-out episodes. In the easier $V0$ scenario, all methods succeed, although central-agent-only PPO follows a substantial detour before recovering a source-directed path. In the more difficult $V2$ scenario, the ring-based policies reach the target in 130 steps and central-agent-only PPO in 272 steps, whereas FCM remains trapped at a local concentration maximum and times out.

For each sensing configuration q at a fixed $v_{\max}$, performance is quantified by the success rate $\text{SR}_q \triangleq \mathbb{P}_q^{\theta_q}(\tau_s \leq$

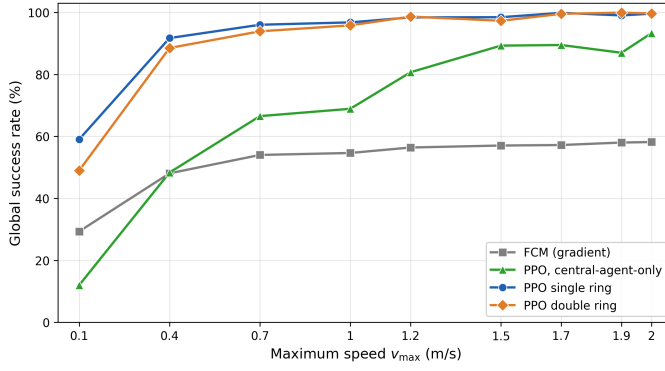
$K_{\max})$ and the mean successful steps $\text{MSS}_q \triangleq \mathbb{E}_q^{\theta_q}[\tau_s \mid \tau_s \leq K_{\max}]$. At $v_{\max} = 0.1$ m/s, the maximum initial source distance compatible with straight-line capture within the finite horizon is $K_{\max} v_{\max} \Delta t + r_s = 1130$ m. Thus, low-speed SR is partly limited by geometric reachability, and MSS must be interpreted only over the successful subset.

Figure 3(a) shows that spatial sensing is particularly valuable at low and intermediate speed limits. At $v_{\max} = 0.4$ m/s, single-ring and double-ring PPO achieve approximately 92% and 89% success, respectively, compared with about 50% for both central-agent-only PPO and FCM. At $v_{\max} = 0.7$ m/s, their success rates increase to approximately 96% and 94%. Central-agent-only PPO generally improves with $v_{\max}$, reaching about 93% at 2 m/s, whereas FCM remains below 60%.

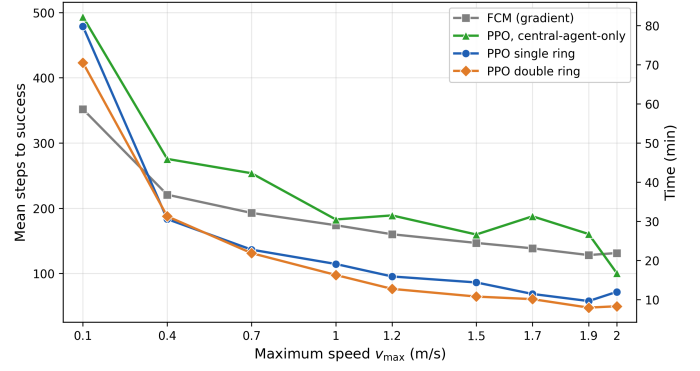
For $v_{\max} \geq 0.7$ m/s, Fig. 3(b) shows that double-ring PPO generally achieves lower MSS than single-ring PPO while maintaining a similar SR. At 1.2 m/s, double-ring PPO requires approximately 75 successful steps, compared with 95 for single-ring PPO, 160 for FCM, and 193 for central-agent-only PPO.

Table III reports the corresponding held-out results at $v_{\max} = 1.2$ m/s by wind-forcing class and initial plume age. Single-ring and double-ring PPO achieve overall success rates of 98.4% and 98.7%, respectively, compared with 79.0% for central-agent-only PPO and 56.4% for FCM. Both ring-based policies remain between 95.4% and 100% across all reported class-wise and age-wise marginals. In contrast, FCM decreases to 29.5% for $V1$ and 34.6% at $\beta = 1/2$, while central-agent-only PPO reaches 64.9% for $V2$ and 68.5% at $\beta = 1/2$. Thus, relative to single-ring PPO, double-ring PPO reduces the conditional mean hitting time by approximately 20% while changing the overall success rate by only 0.3 percentage points.

Within the PPO family, central-agent-only versus single-



(a) Overall success rate (SR).



(b) Mean successful steps (MSS).

Fig. 3. Held-out performance versus $v_{\max}$ for FCM and the three PPO sensing configurations (1560 episodes per point; $\Delta t = 10$ s).

TABLE III
SUCCESS RATE (%) AT $v_{\max} = 1.2$ m/s, OVERALL AND BY WIND-FORCING CLASS AND PLUME AGE.

Method	Overall	V0	V1	V2	V3	$\beta = 1/4$	$\beta = 1/2$	$\beta = 3/4$
FCM	56.4	76.3	29.5	37.2	82.7	79.8	34.6	54.8
PPO central-agent-only	79.0	86.9	86.7	64.9	77.7	91.0	68.5	77.7
PPO single ring	98.4	99.7	100.0	95.6	98.2	100.0	95.8	99.4
PPO double ring	98.7	99.2	100.0	95.4	100.0	100.0	99.6	96.3

ring PPO compares pointwise concentration sensing with the addition of the 20 m ring, whereas single-ring versus double-ring PPO assesses the additional 50 m sensing scale.

A. Limitations and Sim-to-Real Considerations

The evaluation addresses source localization conditional on prior plume detection. Episodes start where the simulated concentration exceeds a prescribed threshold and therefore exclude the time and spatial coverage required to detect the plume from an arbitrary initial position. End-to-end operation will require multi-vehicle exploration and plume detection followed by cooperative source seeking [4], [16].

The present training environment represents a limited subset of the hydrodynamic variability of the study site. All simulations use a single representative open-boundary regime, combined with four prescribed wind-forcing classes. Although this design provides a controlled range of plume responses, it does not assess generalization across distinct circulation regimes. Future work should therefore extend training and evaluation to additional hydrodynamic regimes and jointly consistent wind-marine forcing conditions. Additional processes not represented here, such as wave-induced mixing and circulation, as well as freshwater inflows, should also be considered.

The early-capture component in (7) is proportional to χ_{k+1} . The current policy input z_k^q does not include χ_k . Consequently, the implemented actor cannot explicitly condition its action distribution on the time remaining in the episode. Future work will investigate time-aware policies obtained by including χ_k in the policy input.

The model neglects central-agent inertia, heading and actuator dynamics, and current-induced drift, while assuming

exact ring offsets, noise-free instantaneous measurements, and ideal localization and communication, without collision or energy constraints. Deployment will require domain randomization for sensing and communication uncertainty, autonomous onboard goal detection [15], and progressive software-, hardware-in-the-loop, controlled-release, and coastal field validation.

V. CONCLUSION

We presented a hydrodynamic-aware, single-agent PPO framework for marine pollutant source seeking using finite-history observations, wind and local-current information, joint heading and speed control, and three centralized concentration-sensing configurations. Across the tested speed range, the ring-based policies achieve higher success rates than central-agent-only PPO and the FCM baseline. At $v_{\max} = 1.2$ m/s, single-ring and double-ring PPO achieve 98.4% and 98.7% success, respectively, compared with 79.0% for central-agent-only PPO and 56.4% for FCM. At this operating point, the outer ring provides little additional success probability but reduces the MSS from approximately 95 to 75 steps.

Future work will extend the framework to end-to-end multi-agent missions integrating domain exploration, plume detection, and cooperative source seeking, and will investigate broader hydrodynamic regimes, time-aware policies, and more realistic vehicle, sensing, and communication models.

REFERENCES

- [1] J. A. Farrell, S. Pang, and W. Li, "Chemical plume tracing via an autonomous underwater vehicle," *IEEE Journal of Oceanic Engineering*, vol. 30, no. 2, pp. 428–442, 2005.

- [2] W. Li, J. A. Farrell, S. Pang, and R. M. Arrieta, "Moth-inspired chemical plume tracing on an autonomous underwater vehicle," *IEEE Transactions on Robotics*, vol. 22, no. 2, pp. 292–307, 2006.
- [3] M. Fahad, Y. Guo, B. Bingham, K. Krasnosky, L. Fitzpatrick, and F. A. Sanabria, "Ocean plume tracking with unmanned surface vehicles: Algorithms and experiments," in *Proceedings of the 13th World Congress on Intelligent Control and Automation (WCICA)*, 2018, pp. 1–6.
- [4] E. Fiorelli, N. E. Leonard, P. Bhatta, D. A. Paley, R. Bachmayer, and D. M. Fratantoni, "Multi-AUV control and adaptive sampling in Monterey Bay," *IEEE Journal of Oceanic Engineering*, vol. 31, no. 4, pp. 935–948, 2006.
- [5] J. M. Soares, A. P. Aguiar, A. M. Pascoal, and A. Martinoli, "An algorithm for formation-based chemical plume tracing using robotic marine vehicles," in *OCEANS 2016 MTS/IEEE Monterey*, Monterey, CA, USA, 2016.
- [6] L. Briñón-Arranz, A. Renzaglia, and L. Schenato, "Multirobot symmetric formations for gradient and Hessian estimation with application to source seeking," *IEEE Transactions on Robotics*, vol. 35, no. 3, pp. 782–789, 2019.
- [7] N. Forti, L. Gao, G. Battistelli, and L. Chisci, "Unknown source in spatially distributed systems: Identifiability analysis and estimation," *Automatica*, vol. 136, p. 110025, 2022.
- [8] T. Zhang, V. Qin, Y. Tang, and N. Li, "Distributed information-based source seeking," *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4749–4767, 2023.
- [9] B. Du, K. Qian, C. Claudel, and D. Sun, "Multiagent online source seeking using bandit algorithm," *IEEE Transactions on Automatic Control*, vol. 68, no. 5, pp. 3147–3154, 2023.
- [10] N. Forti, G. Battistelli, and L. Chisci, "Active source seeking in spatio-temporal fields," in *Proceedings of the 29th International Conference on Information Fusion (FUSION)*, Trondheim, Norway, 2026.
- [11] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv 1707.06347*, 2017.
- [12] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dornmann, "Stable-baselines3: Reliable reinforcement learning implementations," *Journal of Machine Learning Research*, vol. 22, no. 268, pp. 1–8, 2021.
- [13] Y. Zhao, B. Chen, X. Wang, Z. Zhu, Y. Wang, G. Cheng, R. Wang, R. Wang, M. He, and Y. Liu, "A deep reinforcement learning based searching method for source localization," *Information Sciences*, vol. 588, pp. 67–81, 2022.
- [14] M. Park, P. Ladosz, and H. Oh, "Source term estimation using deep reinforcement learning with gaussian mixture model feature extraction for mobile sensors," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 8323–8330, 2022.
- [15] Y. Shi, M. Wen, Q. Zhang, W. Zhang, C. Liu, and W. Liu, "Autonomous goal detection and cessation in reinforcement learning: A case study on source term estimation," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 1, 2025, pp. 738–745.
- [16] E. Sebastián, T. Duong, N. Atanasov, E. Montijano, and C. Sagüés, "Physics-informed multi-agent reinforcement learning for distributed multi-robot problems," *IEEE Transactions on Robotics*, 2025.
- [17] L. P. Kaelbling, M. L. Littman, and A. R. Cassandra, "Planning and acting in partially observable stochastic domains," *Artificial Intelligence*, vol. 101, no. 1–2, pp. 99–134, 1998.
- [18] DHI, *MIKE 21 Flow Model FM: Hydrodynamic Module*, Scientific Documentation, Danish Hydraulic Institute, Hørsholm, Denmark, 2023.
- [19] E. Clementi, M. Drudi, A. Aydogdu, A. Moulin, A. Grandi, A. Mariani, A. C. Goglio, J. Pistoia, P. Miraglio, R. Lecci, F. Palermo, G. Coppini, S. Masina, and N. Pinardi, "Mediterranean sea physical analysis and forecast (CMEMS MED-physics, EAS8 system)," Data set, 2023.
- [20] P. Oddo and N. Pinardi, "Lateral open boundary conditions for nested limited area models: A scale selective approach," *Ocean Modelling*, vol. 20, pp. 134–156, 2008.
- [21] H. Hersbach *et al.*, "The ERA5 global reanalysis," *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020.
- [22] J. Schulman, P. Moritz, S. Levine, M. I. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," in *International Conference on Learning Representations (ICLR)*, 2016.
- [23] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proceedings of the 3rd International Conference on Learning Representations (ICLR)*, 2015.